P.PORTO

ESCOLA
SUPERIOR
DE MEDIA
ARTES
E DESIGN
POLITÉCNICO
DO PORTO

# ALP

FICHA DE EXERCÍCIOS
ATIVIDADE LETIVA

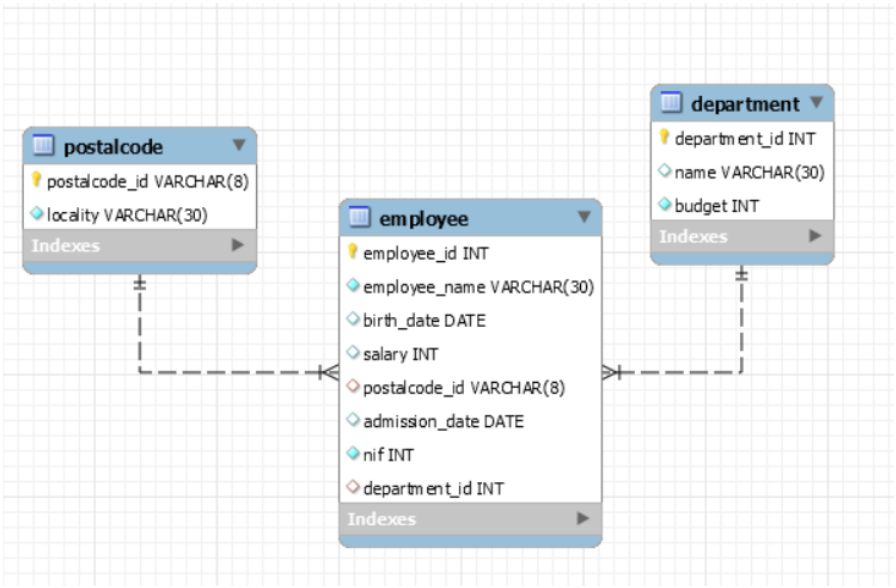
**Databases**

UNIDADE CURRICULAR

**SQL Language**

FICHA

Consider the database ***Company***, implemented in the previous exercise sheet, with the following scheme:




Construct the SQL Queries to obtain the following results:

1. Number of employees for each department

Result Grid | Filter Rows:

| number | department_id |
|---|---|
| 3 | 1 |
| 1 | 2 |
| 2 | 3 |
| 2 | 4 |
| 2 | 5 |

2. Number of employees for each department (name of department)

| | COUNT(*) | name |
|---|---|---|
| ▶ | 3 | Produção |
| | 1 | Contabilidade |
| | 2 | Projetos |
| | 2 | Recursos Humanos |
| | 2 | Logística |

3. Average salary by department

| | AVG(salary) | department_id |
|---|---|---|
| ▶ | 1640.0000 | 1 |
| | 1950.0000 | 2 |
| | 2100.0000 | 3 |
| | 2250.0000 | 4 |
| | 1420.0000 | 5 |

4. Average salary by department (name of department)

| | AVG(salary) | name |
|---|---|---|
| ▶ | 1640.0000 | Produção |
| | 1950.0000 | Contabilidade |
| | 2100.0000 | Projetos |
| | 2250.0000 | Recursos Humanos |
| | 1420.0000 | Logística |

5. Id and name of each employee, followed by the designation of the department

| | employee_id | employee_name | name |
|---|---|---|---|
| ▶ | 1 | Manuel Santos | Produção |
| | 2 | Paulo Fosneca | Produção |
| | 3 | Carla Carolina | Produção |
| | 5 | Maria Costa | Contabilidade |
| | 4 | Isabel Antunes | Projetos |
| | 10 | Diogo Dionísio | Projetos |

6. Id and name of the employees in the *Contabilidade* department

| | employee_id | employee_name | name |
|---|---|---|---|
| ▶ | 5 | Maria Costa | Contabilidade |

7. List of employees (id, name, nif, postalcode, locality) sorted by location and sub-ordinate by employee name

| | employee_id | employee_name | nif | postalcode_id | locality |
|---|---|---|---|---|---|
| ▶ | 10 | Diogo Dionísio | 213456789 | 4460-100 | Matosinhos |
| | 9 | Liliana Lousada | 901234567 | 4460-100 | Matosinhos |
| | 3 | Carla Carolina | 345678901 | 4100-050 | Porto |
| | 7 | José Silva | 789012345 | 4100-050 | Porto |
| | 1 | Manuel Santos | 123456789 | 4000-100 | Porto |
| | 8 | Maria Andrade | 890123456 | 4000-100 | Porto |
| | 2 | Paulo Fosneca | 234567890 | 4000-100 | Porto |

8. Total salaries by department (id and department designation)

| department_id | name | SUM(salary) |
|---|---|---|
| 1 | Produção | 4920 |
| 2 | Contabilidade | 1950 |
| 3 | Projetos | 4200 |
| 4 | Recursos Humanos | 4500 |
| 5 | Logística | 2840 |

9. List number of employees in the department *Produção*

| num_employes |
|---|
| 3 |

10. List employee data (id, name, postalcode and location, including the code and designation of each department; list sorted by department.

| employee_id | employee_name | postalcode_id | locality | department_id | name |
|---|---|---|---|---|---|
| 5 | Maria Costa | 4480-876 | Vila do Conde | 2 | Contabilidade |
| 7 | José Silva | 4100-050 | Porto | 5 | Logística |
| 8 | Maria Andrade | 4000-100 | Porto | 5 | Logística |
| 1 | Manuel Santos | 4000-100 | Porto | 1 | Produção |
| 2 | Paulo Fosneca | 4000-100 | Porto | 1 | Produção |
| 3 | Carla Carolina | 4100-050 | Porto | 1 | Produção |

11. List sum of the amount payable to Social Security (11% of salary), by department (departmental designation)

| payment | name |
|---|---|
| 541.20 | Produção |
| 214.50 | Contabilidade |
| 462.00 | Projetos |
| 495.00 | Recursos Humanos |
| 312.40 | Logística |

12. List of employees (id, name and nif) of the *Produção* and *Recursos Humanos* departments

| employee_id | employee_name | nif | name |
|---|---|---|---|
| 1 | Manuel Santos | 123456789 | Produção |
| 2 | Paulo Fosneca | 234567890 | Produção |
| 3 | Carla Carolina | 345678901 | Produção |
| 6 | Ricardo Rocha | 678901234 | Recursos Humanos |
| 9 | Liliana Lousada | 901234567 | Recursos Humanos |